MILITARY SPECIFICATION

BOARD ASSEMBLY, COMPONENT 11748254

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers component board assembly used in Power Supply Subassembly, PP6324/TPM-22, part of the Radar Test Set AN/TPM-22.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

F11748254	Board Assembly, Component
B11740249	Wire Runs List
C11740251	Schematic, Component Board Assembly
B11748264	"T" Connector
C11748263	Hi Voltage Cable

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PACKAGING DATA SHEET

F11748254 Packaging of Component Board Assembly

(Copies of specifications, standards, drawings and packaging data sheets required by the suppliers in connection with specific functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The component board shall be manufactured in accordance with Drawing F11748254 and drawings pertaining thereto.

3.2 General specifications.- Where applicable, the component board shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration.- The component board shall be mounted on the vibration table in normal service position. The component board shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 35 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock.- The component board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 $\pm$ 1 milliseconds (ms). The G value of each impulse shall be within $\pm$ 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

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3.3.3 Altitude.

3.3.3.1 Operating.- The component board shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating.- The component board shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating.- The component board shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

3.3.4.2 Nonoperating (storage).- The component board shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65°F and +160°F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating).- The component board shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 ± 5 percent for a period of not less than 10 days.

3.4 Electrical.

3.4.1 Continuity.- Each conductor shall be continuous electrically as specified on the schematic diagram C11740251, and wire runs list B11740249. The wire runs list shall take precedence over the schematic.

3.5 Performance.

3.5.1 Relay K1.- With 24V±0.5Vdc applied between TB6-4(+)[1] and TB6-5(-), the resistance between terminal W3 and TB6-5 shall be 1.0M ± 10%. With no voltage applied, the resistance shall be open-circuited.

3.5.2 Relay K2.- With 24V ± 0.5Vdc applied between TB6-3(+) and TB6-5(-), the resistance between terminal W2 and TB6-1 shall be open-circuited. With no voltage applied, the resistance shall be 1.1 megohms ± 10%.

3.5.3 1K Resistor.- The resistance measured between TB6-1 and TB6-5 shall be 1.0 kilohm ± 10%.

3.5.4 Spark Gap (SP1).- With a variable DC voltage applied between TB6-1(+) and TB6-5(-), spark gap (SP1) shall break down at 150Vdc ± 5%.

1 The information in parenthesis refers to the polarity applied to this terminal. This notation shall be used throughout this specification.

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3.5.5 10KV Test.- With 10 KVDC ± 100V applied between terminals W2 and W3(+), and TB6-5(-); the voltage measured between TB6-1(+) and TB6-5(-) shall be 9.1Vdc ± 5%. When 24 ± 0.5Vdc is applied between TB6-3(+) and TB6-5(-), the 9.1V between TB6-1(+) and TB6-5(-) shall fall to 0 ± 0.4Vdc. When 24 ± 0.5Vdc is applied between TB6-4(+) and TB6-5(-) the HI-VOLTAGE power supply output current shall increase by 100 milliamps ± 10%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) submission.- The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three component boards manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions

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of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests.- The requirements and tests in Table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60°F and 90°F).

TABLE I - CLASSIFICATION OF DEFECTS
(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR:		
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3
MINOR: NONE		

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- One component board shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table III. Three component boards shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table IV.

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TABLE II - SPECIAL SAMPLING

	CONTROL	REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings - Visual
302.	General specifications	3.2	MIL-F-13926 & MIL-C-46357 - Visual

TABLE III - SPECIAL SAMPLING, ENVIRONMENTAL

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude	3.3.3	4.6.1.3
306.	Temperature	3.3.4	4.6.1.4
307.	Humidity (nonoperating)	3.3.5	4.6.1.5

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment.- Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of

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a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment.- In conjunction with 4.5.2.2, the standard equipment listed in Table IV shall be utilized in performing the applicable tests of 4.6.

TABLE IV - TEST EQUIPMENT LIST

ITEM		DESCRIPTION
1.	Power Supply	24Vdc, @ 5A min., regulated to ± 2%
2.	Digital Voltmeter	DC, 0 to 200V, Accuracy = ± 0.1% of reading. Input impedence of 10 megohm min.
3.	Megger	100 to 500Vdc ± 10%, 0 to 100 megohms min., accuracy = ± 10%
4.	Power Supply	0 to 200Vdc, @ 100 MA min., variable accuracy = ± 1%
5.	Power Supply	0 to 15KVdc, @ 300 MA, Accuracy = ± 1%
6.	"T" Connector	Per Frankford Arsenal B11748264
7.	Switches S1 & S2	Grayhill Inc., Part No. 35-1 (N.O.)
8.	Connector J1	Per MIL-C-39024/10-12
9.	HI-Voltage Cable	Per Frankford Arsenal C11748263-1

4.6.1 Environmental.- Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration.- The test shall be performed as specified in Method 514, Procedure IX, Part I of MIL-STD-810, except that failure criteria do not apply. After vibration, the component board shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1.

4.6.1.2 Shock.- This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 G. After the shock impulses, the component board shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

4.6.1.3 Altitude (nonoperating and operating).- This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 feet

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(5.5 in Hg). Operation of Step 3 shall be the performance tests specified in 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating).- This test shall be performed as specified in Procedure I, Method 501 of MIL-STD-810. The temperature of Step 2 shall be +160°F. The temperature of Step 3 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be +155°F. In Step 5 the temperature shall be maintained for 4 hours and the component board shall be tested in accordance with 4.6.2 and 4.6.3 during this time.

4.6.1.4.2 Low (operating and nonoperating).- This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -65°F maintained for not less than 4 hours. The temperature of Step 4 shall be -40°F. In Steps 5 and 7, the component board shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity (nonoperating).- This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The component board shall be tested in accordance with 4.6.3.

4.6.2 Electrical.- Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance.- The selected component boards shall be tested as follows:

4.6.3.1 Test circuit.- Using applicable items designated in Table IV, construct the test circuit illustrated in Figure 1. Initially the power supplies shall be turned off. The following test procedures refer to this test circuit.

4.6.3.2 Relay K1.- Disconnect W3 from the "T" connector. Connect the megger, item 3 of Table IV, between W3 and common terminal TB6-5. Adjust megger to 500V operation. Verify the open-circuit resistance requirement of 3.5.1. Turn on +24V power supply, item 1, and depress switch S1. Verify the 1.0M resistance requirement of 3.5.1. Turn off the +24V power supply, disconnect the megger, and reconnect W3 to the "T" connector.

4.6.3.3 Relay K2.- Disconnect W2 from the "T" connector. Connect the megger, item 3 of Table IV, between W2 and terminal J1. Adjust megger to 500V operation. Verify the 1.1 M resistance requirement of 3.5.2. Turn on +24V power supply, item 1, and depress switch S2. Verify the open circuit requirement of 3.5.2. Turn off the +24V power supply, disconnect the megger, and reconnect W2 to the "T" connector.

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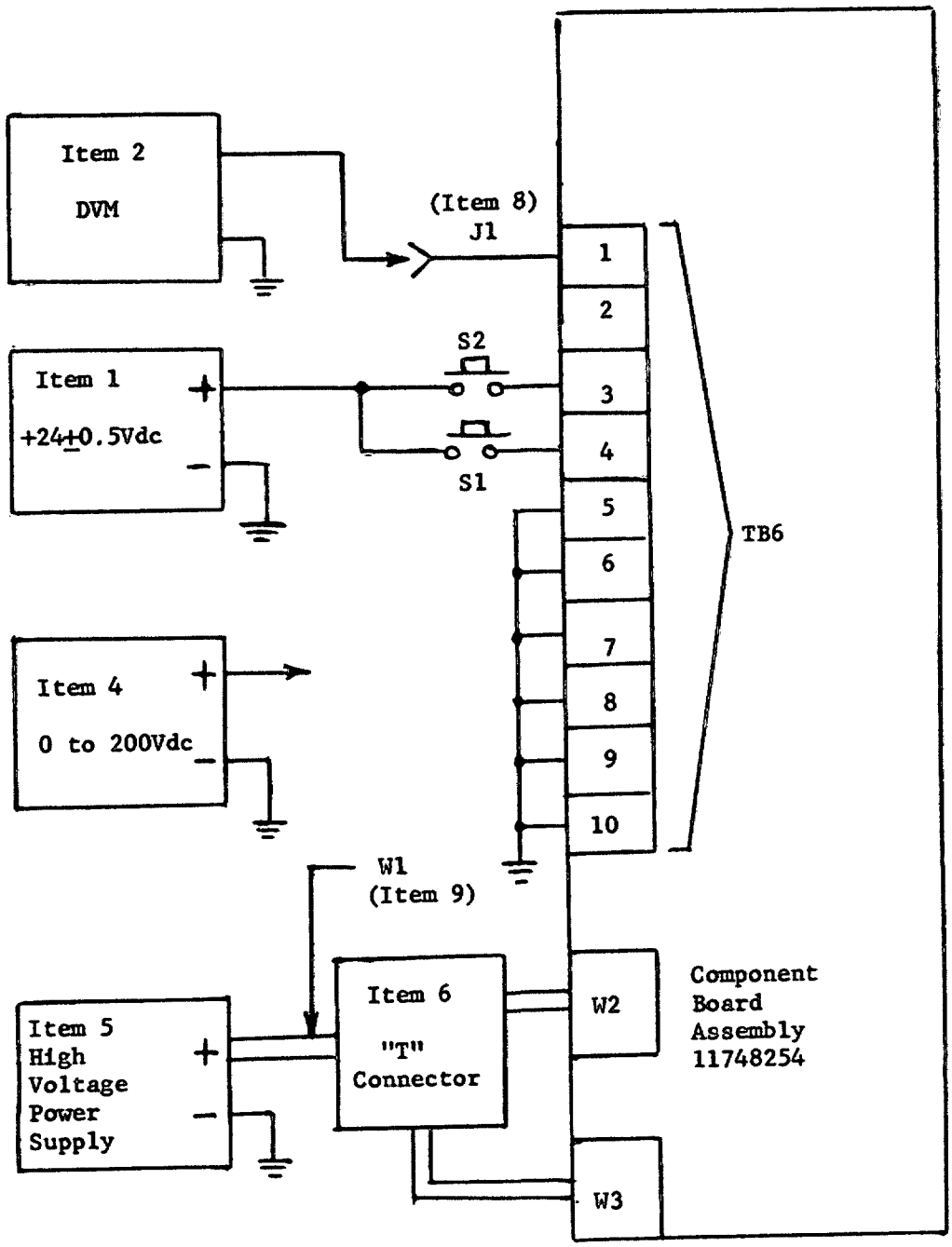


FIGURE 1. COMPONENT BOARD ASSEMBLY, TEST CIRCUIT

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4.6.3.4 1K Resistor.- Connect the megger, item 3 of Table IV, between terminal J1 and common terminal TB6-5. Adjust megger to 100V operation. Verify the resistance requirements of 3.5.3. Disconnect the megger.

4.6.3.5 Spark gap (SP1).- Connect variable power supply, item IV to J1(+) and TB6-5(-). Connect the digital voltmeter, item 2 of Table IV across the same terminals. Vary the DC power supply and verify the spark gap breakdown requirements of 3.5.4. Disconnect the voltmeter and power supply.

4.6.3.6 10KV test.- Turn on high voltage power supply, item 5 of Table IV, and adjust to 10 KVdc ± 100V. Connect the digital voltmeter, item 2 of Table IV, to J1(+) and common terminal TB6-5(-). Verify the 9.1Vdc requirement of 3.5.5 at terminal J1. Turn on the +24Vdc power supply, item 1 of Table IV, and depress switch S2. Verify the zero volt requirement of 3.5.5 at terminal J1. Release switch S2, and depress switch S1. Verify the increase in high voltage current as specified in 3.5.5. Turn off the high voltage, and +24Vdc supplies.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11748254. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The component board assembly is used in Power Supply Subassembly, PP-6324/TPM-22, part of Radar Test Set AN/TPM-22. The component board assembly is used to monitor the characteristics of the 10KV voltage present in Radar Set AN/VPS-2.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable packaging data sheet number (see 5.1).
c. Selection of applicable level of preservation packaging and packing.
d. Level of protection required.
e. Applicable stock number.
f. Requirements for first article submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 4931-A147

☆U.S. GOVERNMENT PRINTING OFFICE: 1976-603-766/3454

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-B-48594(MU) Board Assembly, Component 11748254

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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